

GENERAL SPECIFICATION G5.23

Lightning Protection System

CONTENTS

G5.23	Lightning Protection System	1
G5.23.1	General.....	1
G5.23.2	Products	3
G5.23.3	Execution.....	4
G5.23.4	Lightning Warning System	7

G5.23 LIGHTNING PROTECTION SYSTEM

G5.23.1 General

(a) Introduction

- (i) This Section of the specifications is intended to cover the design, supply, deliver, installation, testing and commissioning of the lightning protection system for all structures constructed under the Contract.
- (ii) The lightning protection system shall be designed and installed in strict accordance with SS 555 and the satisfaction of the S.O..
- (iii) The lightning protection system Standard installation details are only for reference. In accordance with SS 555, Contractor shall carry out design in accordance with General Principles, Risk Management, and provide Protection against lightning for physical damage to structures, life hazard, and electrical and electronic systems within structures, and endorsed by LEW, with appropriate Engineer Grade. The complete lightning protection system shall be installed by the competent experienced installers.

(b) References

- (i) The following is a list of standards which may be referenced in this Section:

1) British Standards Institute (BSI)

Standard	Subject
BS EN 12385	Stranded Steel Wire Ropes.
BS 1449	Steel Plate, Sheet and Strip. Carbon and Carbon-Manganese Plate, Sheet and Strip.
BS 6004	Specification for PVC-Insulated Cables (Non-Armoured) for Electric Power and Lighting.
BS 7884	Copper and Copper-Cadmium Stranded for Overhead Electric Traction and Power Transmission Systems.
BS EN 1057	Copper and Copper Alloys. Seamless, Round Copper Tubes for Water and Gas in Sanitary and Heating Applications.
BS EN 1652	Copper and Copper Alloys. Plate, Sheet, Strip and Circles for General Purposes.
BS EN 12163	Copper and Copper Alloys. Rod for General Purposes.
BS EN 12164	Copper and Copper Alloys. Rod for Free Machining Purposes.
BS EN 12165	Copper and Copper Alloys. Wrought and Unwrought Forging Stock.

Standard	Subject
BS EN 12167	Copper and Copper Alloys. Profiles and Rectangular Bar for General Purposes.
BS EN 12420	Copper and Copper Alloys. Forgings.

2) International Electrotechnical Commission (IEC)

Standard	Subject
IEC 60060	High Voltage Test Techniques
IEC 60079	Explosive Atmospheres
IEC 60099	Surge Arresters
IEC 60228	Conductors of insulated cables
IEC 60245	Rubber Insulated Cables of Rated Voltages Up to and Including 450/750V
IEC 60364	Electrical Installations of Buildings
IEC 60502	Power cables with extruded insulation and their accessories for rated voltages from 1 kV (Um = 1,2 kV) up to 30 kV (Um = 36 kV)
IEC 61643	Surge Protective Devices Connected to Low-Voltage Power Distribution Systems
IEC 62305	Protection against lightning

3) Spring Singapore (SS)

Standard	Subject
CP 5	Code of Practice for Electrical Installation
SS 551	Code of Practice for Earthing
SS 555	Code of Practice for protection against lightning
SS 322	Earthing and Bonding Clamps
SS 358	Polyvinyl Chloride Insulated Cables of Rated Voltages Up to and Including 450/750 V

(c) Submittals

(i) Shop Drawings, Technical Data: As a minimum, submit the following items:

- 1) Make, model of complete lightning system components;
- 2) Lightning protection system layout, cross-sections, and bonding details;
- 3) Installation method drawings;
- 4) Risk Assessment and mitigation of risk in accordance with SS 555;

- 5) Field test reports;
 - 6) Earthing resistance test witness certification; and
 - 7) BCA's Certification of Supervision and Proper Installation of Lightning Protection System.
- (d) Quality Assurance
- (i) Manufacturer: All system components shall be the product of a manufacturer regularly engaged in the manufacturing of lightning protection components in accordance with SS or IEC or BS standard.
 - (ii) Installer: Lightning protection system shall be installed under the direct supervision of the Contractor's Licensed Electrical Worker (LEW):
 - 1) Certification: Final installation and earthing connection inspection of the System shall be certified by LEW. BCA's Certification of Supervision of Lightning Protection System shall be certified by LEW (Engineer grade).
 - (iii) Compliance: Materials manufactured shall conform to SS or IEC or BS.

G5.23.2 Products

- (a) General
- (i) System Material: High copper content, tinned copper, unless otherwise specified; and
 - (ii) All material shall comply in weight, size, and composition for the class of structure to be protected as established by SS 555.
- (b) Air Terminal
- (i) General: Air terminations shall be by means of a network of horizontal conductors installed on the roof; and
 - (ii) Material: 25 mm x 3 mm soft annealed and high conductivity tinned copper tape.
- (c) Down Conductors
- (i) Material: 25 mm x 3 mm tinned copper tape, and/or building reinforcement steel structure columns, provided they are electrically continuous and adequately earthed, as appropriate for application;
 - (ii) Provide stainless steel test link boxes with test clamp mounted 1,500 mm above ground level for connection to each earth electrode conductor via 25 mm x 3 mm tinned copper tape;
 - (iii) Copper Cable: Bare medium hard-drawn stranded, having 97.5 percent minimum conductivity; and
 - (iv) Bonding Conductor: Flexible tinned copper strap, minimum 25 mm wide x 3 mm thick.
- (d) Earth Electrode Conductor and Inspection Pit
- (i) Material: Individual 16 mm diameter copper-clad steel rod with a radial thickness of 0.25 mm of 99.9 percent pure electrolytic copper molecularly bonded onto a steel core. Coupling shall be of long length high copper alloy.

- (ii) Inspection pit shall be concrete or extremely high corrosion resistance and of long life span, UV-stable, shatter-proof, chemical-resistant polymer earth pit of approved design. Capable of withstanding the vehicle load of 10,000 kg for pit located at driveway and 5,000 for pit located at non-driveway.
 - (iii) Cover shall be permanently marked EARTH ELECTRODE INSPECTION PIT.
- (e) Conductor Fastener and accessories
 - (i) Down conductors shall be connected to roof tapes by means of end or ridge air termination clamping device with bi-metallic joints as appropriate; and
 - (ii) Capable of withstanding minimum pull of 45 kilograms.
- (f) Fittings
 - (i) Heavy-duty; and
 - (ii) Bolts, Screws, and Related Hardware: Stainless steel.
- (g) Earthing Connections
 - (i) Welds: Exothermic process, “Thermoweld”, “Cadweld” or equal;
 - (ii) Fasteners: Bolted clamp type, corrosion-resistant copper alloy; and
 - (iii) Hardware: Copper.
- (h) Conductor Connections and Splicers
 - (i) Welds: Exothermic process, “Thermoweld”, “Cadweld” or equal; and
 - (ii) Fasteners: Bolted clamp type, corrosion-resistant copper alloy as appropriate for application.
- (i) Conduit
 - (i) UPVC, as specified in General Specification G5.11 – Raceways.

G5.23.3 Execution

- (a) General
 - (i) Installation of bare copper materials on aluminium surfaces will not be permitted;
 - (ii) Provide pitch pockets and waterproof all roof penetrations;
 - (iii) Install system in inconspicuous manner so that components blend with building aesthetics;
 - (iv) Waterproof all joints between dissimilar materials to prevent corrosion; and
 - (v) Provide identification of labels engraved on the earth pit cover along with final measured resistance value in ohms.
- (b) Inspection
 - (i) Verify conditions prior to installation. Actual conditions may require adjustments in air terminal and earth rod locations;

- (c) Air Terminals
 - (i) Secure base to roof surface with clamps, tape clip, slate holdfast and ridge saddle as appropriate for application at an interval not exceeding 900 mm;
 - (ii) Provide terminal flashing at roof penetrations;
 - (iii) Provide copper strike pads between air terminals and down conductors for pitch tile roof; and
 - (iv) Mesh Network Maximum Spacing: 10 m x 20 m; and
 - (v) The method of clamping for the roof terminations shall be such that the waterproofing of the roof surface, roof or any part of the roof materials that has been water proofed shall be not punctured.
- (d) Down Conductors
 - (i) To be concealed;
 - (ii) Provide 40 mm UPVC conduit in building walls or columns for main down conductors where practical;
 - (iii) Conductor Maximum Spacing: 30 m;
 - (iv) Support: Secure with brass DC tape clips, maximum spacing for:
 - 1) Vertical: 1,000 mm; and
 - 2) Horizontal: 1,000 mm.
 - (v) Maintain horizontal and vertical conductor courses free from dips or pockets; and
 - (vi) Bends: Maximum 90 degrees, with minimum 200 mm radius.
- (e) Bonding
 - (i) Bond to Air Terminal or Down Conductor System
 - 1) All roof-mounted ventilators, fans, air handlers, masts, flues, cooling towers, handrails, water tanks, pumps, metallic pipes, antenna and other sizeable metal objects; and
 - 2) Roof flashing, gravel stops, insulation vents, ridge vents, roof drains, soil pipe vents, and other small metal objects and another earthed objects.
 - (ii) Bond each steel column or major framing members to earthing system;
 - (iii) Bond each down conductor to earth electrode system; and
 - (iv) Bond each down conductor to building reinforcing steel in structural columns via earth test link box.
- (f) Earth Electrode System
 - (i) Earth Termination Network Conductor
 - 1) Completely encircle building structure;
 - 2) Bury minimum 900 mm below finished grade; and
 - 3) Minimum 1,000 mm distance from foundation walls.

- (ii) Interconnect earth rods by tape clamps, direct-buried copper conductors;
- (iii) Minimum Length per Electrode per Pit : 3,600 mm;
- (iv) Each earth electrode shall be protected by an inspection pit;
- (v) Earth electrode shall be 16 mm in diameter; and
- (vi) Maximum Resistance: 10 ohms when connected to earth rods.
- (vii) Connections
 - 1) Install earth electrode conductors continuous between connections;
 - 2) Exothermic welded connections to structural steel, and buried and non-accessible connections;
 - 3) Provide bolted clamp type mechanical connectors for all exposed connections;
 - 4) Provide interconnections with electrical and telephone systems and all underground water, gas, and sewer metal pipes; and
 - 5) Provide electric service arrestor earth wire to building water mains.
- (viii) Earth Potential Equalisation
 - 1) Provide surge diverter to connect independent “clean” earthing system, main earth terminal and lightning earth terminal networks;
 - 2) Such component shall be robust, housing conform to IP 55;
 - 3) Where buried it shall be protected by an inspection pit; and
 - 4) Rating
 - 100 kA 8/20s surge rating; and
 - 230V “breakdown” voltage.
- (g) Field Testing
 - (i) Isolate lightning protection system from other earth conditions while performing tests; and
 - (ii) Resistance: Test earth resistance of earthing system by the fall-of-potential method:
 - 1) Test Resistance to earth: Maximum 10 ohms; and
 - 2) Install additional earth electrodes as required to obtain allowable resistance value.
 - (iii) Test Report
 - 1) Description of equipment tested;
 - 2) Description of test;
 - 3) Test results;
 - 4) Conclusions and recommendations;
 - 5) Include appropriate test forms as supplements to the report;
 - 6) Identification of test equipment used;

- 7) Signature of responsible test organisation authority; and
- 8) Earth resistance test for the lightning protection system shall be carried out in the presence of the S.O. or the representative of the S.O and the Contractor's LEW, who designed the lightning protection system. Earth resistance test report complete with layout diagram shall be certified by the same LEW, while making official submissions.

G5.23.4 Lightning Warning System

- (a) General
 - (i) The lightning warning system shall be used to detect lightning activity and provide a warning to the facility's personnel of the lightning risks.
- (b) System Architecture and Components
 - (i) Sensor Assembly
 - 1) The sensor assembly shall be omnidirectional and able to sense lightning strikes and significant changes in the electric field defined in Table G5.23-1 and Table G5.23-2;
 - 2) The sensor assembly shall be provided with a battery backup of at least two hours of operation in the event of normal supply failure; and
 - 3) The sensor assembly shall be suitable for installation on a mast at a flat ground and at locations agreed by the equipment manufacturer and approved by S.O..

Table G5.23-1: Distances of Lightning Strikes

Detection of Lightning Strikes	Distance
Far Strikes	Approx. 10 – 25 km
Near Strikes	Less than 10 km

Table G5.23-2: E-field Measurements for Lightning Detection

Lightning E-Field Measurement	Magnitude
Low E-Field	± 3 to 7 kV/m
High E-Field	± 7 to 10 kV/m

- (ii) Console
 - 1) A console shall receive information from the sensor assembly via wireless communication. This shall include the lightning warning levels and any recent lightning related activity such as lightning strikes and electric field levels.
 - 2) The information received from the sensor assembly shall be logged by the console. The console shall be able to log at least 150 events.
 - 3) The console shall be accessible via Ethernet through the use of a web browser and an IP address for any configuration or accessing the logged events.

(iii) SMS (Short Messaging Service) Communication Modules

- 1) A master SMS communication module shall be located near the console and receive the lightning warning from the console. Slave SMS modules shall be installed at various locations of the facility as specified in the Particular Specifications.
- 2) The master and slave communication modules shall be provided with a battery backup of at least two hours of operation in the event of normal supply failure. The battery bank shall be housed in a separate enclosure from the master and slave modules. The location of this enclosure shall be as close as possible to the master and slave modules' enclosures and weatherproof type where installed outdoors.
- 3) Sending of lightning warnings between the master and slave SMS communication modules shall be via 3G or 4G service.
- 4) The master module shall be capable of storing at least 24 phone number slots (including the slaves') and capable of sending SMS notifications to interchangeable users.

(iv) Siren and Beacon

- 1) A set of sirens and beacon shall be provided at the location of each master and slave SMS communication modules.
- 2) The siren shall sound for the two following events:
 - Detection of near strike or high e-field; and
 - All clear status.
- 3) The sound for the two events shall be unique to differentiate the status of the lightning activity and shall have an adjustable duration. The sirens shall have a loudness of at least 105 db.
- 4) The beacon shall be amber and LED type that is be activated upon detection of near strike of high e-field and is only deactivated upon an all clear status.

(c) System Performance

- (i) The system shall be fully automated and no human intervention is required for the safe and complete operation of the lightning warning system.
- (ii) Users shall not be allowed to adjust parameters of the lightning strike nor the e-field measurements.

(d) Other Requirements

- (i) Installation of the lightning warning system shall comply with the requirements of the equipment supplier.
- (ii) The Contractor shall provide the license, any authentication key or manuals that may come with the system at the end of the Contract.
- (iii) The system shall be protected against lightning induced over voltages in accordance with local codes. All metal frames of the enclosures shall be effectively earthed and incoming power supply shall be protected by surge arrestors.

- (iv) All equipment shall be held firmly in place. This shall include any mounts, fastenings and supports adequate and in accordance with the equipment supplier's requirements. The enclosures and housing for the equipment shall suit the installation location.
- (v) The Contractor shall test the system to show that its performance satisfies the requirement of this specifications in accordance with the equipment supplier's recommendations. Detailed test procedures shall be provided for commissioning.
- (vi) The Contractor shall provide training of the operations personnel.

END OF SECTION